

Homework 2 – Set identities

Session 2 · Sets & set operations

HOW THIS IS MARKED

Every answer carries one sentence of justification. A correct value with no reasoning scores half. A wrong value with a clear, nearly-correct argument scores most of the marks. You may discuss problems with anyone and may use AI tools – but you must be able to explain every line you submit.

TWO FORMATS, AND THEY ARE MARKED STRICTLY

To prove an identity: double inclusion. Take an arbitrary element of one side, show it lies in the other, then do the reverse. The word "arbitrary" must appear.

To disprove one: name your sets explicitly, compute both sides, and point at the element that is in one and not the other. A Venn diagram is not a proof and is not a disproof.

Part A · Notation and computation (6 marks)

1 NOTATION

2 marks

Let $U = \{1, 2, \dots, 12\}$, $A = \{x \in U : x \text{ is even}\}$ and $B = \{x \in U : x \text{ is a multiple of } 3\}$.

- Write A and B in roster notation.
- Compute $A \cap B$, $A \cup B$, $A \setminus B$, $A \triangle B$ and B^c .

SOLUTION

(a) $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12\}$.
 (b) $A \cap B = \{6, 12\}$; $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$; $A \setminus B = \{2, 4, 8, 10\}$;
 $A \triangle B = \{2, 3, 4, 8, 9, 10\}$; $B^c = \{1, 2, 4, 5, 7, 8, 10, 11\}$. 1 mark for (a), 1 for (b); lose the (b) mark if B^c is given without reference to U .

2 $\in$ VS $\subseteq$

2 marks

Let $C = \{1, \{1\}, \{1, 2\}, \emptyset\}$. Decide true or false, and in each case quote the definition that settles it.

- $1 \in C$ (b) $\{1\} \in C$ (c) $\{1\} \subseteq C$ (d) $\{1, 2\} \subseteq C$ (e) $\emptyset \subseteq C$ (f) $|C| = 4$

SOLUTION

(a) True – 1 is listed. (b) True – $\{1\}$ is listed as an element. (c) True – the only element of $\{1\}$ is 1, and $1 \in C$. (d) False – it would need $2 \in C$, and 2 is not an element of C . (e) True – vacuously, for every set. (f) True. 1 mark for (a)–(c), 1 for (d)–(f). Note (b) and (c) are both true here for different reasons; a student who says "they cannot both be true" has the misconception this question is testing.

3 POWER SET

2 marks

Let $A = \{a, b, c\}$.

1. List $\mathcal{P}(A)$ in full, organised by the size of the subset.
2. State $|\mathcal{P}(A)|$ and explain in one sentence why $|\mathcal{P}(A)| = 2^{|A|}$ for *any* finite A .

SOLUTION (a) $\emptyset; \{a\}, \{b\}, \{c\}; \{a, b\}, \{a, c\}, \{b, c\}; \{a, b, c\}$.

(b) $|\mathcal{P}(A)| = 8 = 2^3$. Building a subset means deciding, independently for each of the n elements, whether it is in or out – two choices n times, so 2^n subsets. 1 mark for a complete list (dropping $\emptyset$ or A itself loses it), 1 for the choice argument. "Because the formula says so" scores zero.

Part B · Six identities to prove (12 marks)

FOR EVERY PROBLEM IN THIS PART Prove by double inclusion. Each is 2 marks: 1 for each direction. A proof that verifies the identity on one example scores **zero**, however carefully the example is computed.

4 DISTRIBUTIVITY

2 marks

Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for all sets A, B, C .

SOLUTION ($\subseteq$) Let $x \in A \cup (B \cap C)$ be arbitrary. Then $x \in A$, or $x \in B \cap C$. If $x \in A$ then x is in both $A \cup B$ and $A \cup C$, so in their intersection. If $x \in B \cap C$ then $x \in B$ and $x \in C$, so again $x \in A \cup B$ and $x \in A \cup C$. Either way $x \in (A \cup B) \cap (A \cup C)$. ($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$ be arbitrary, so $x \in A \cup B$ and $x \in A \cup C$. If $x \in A$ we are done. If not, then from the first $x \in B$ and from the second $x \in C$, so $x \in B \cap C$. Either way $x \in A \cup (B \cap C)$. ■ The "if not, then" case split in the second direction is where the marks are lost.

5 DE MORGAN

2 marks

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$.

SOLUTION ($\subseteq$) Let $x \in (A \cap B)^c$ be arbitrary. Then $x \in U$ and $x \notin A \cap B$, i.e. it is not the case that both $x \in A$ and $x \in B$. So $x \notin A$ or $x \notin B$, giving $x \in A^c$ or $x \in B^c$, hence $x \in A^c \cup B^c$. ($\supseteq$) Let $x \in A^c \cup B^c$ be arbitrary. Then $x \notin A$ or $x \notin B$, so it is not the case that x lies in both, i.e. $x \notin A \cap B$. As $x \in U$, $x \in (A \cap B)^c$. ■ 1 mark per direction. The negation of "and" into "or" must be stated, not assumed silently.

6 DIFFERENCE

2 marks

Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ for all sets A, B, C .

SOLUTION ($\subseteq$) Let $x \in A \setminus (B \cup C)$ be arbitrary. Then $x \in A$ and $x \notin B \cup C$, so $x \notin B$ and $x \notin C$. Hence $x \in A \setminus B$ and $x \in A \setminus C$, so x is in the intersection.
 ($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$ be arbitrary. Then $x \in A$, $x \notin B$ and $x \notin C$, so $x \notin B \cup C$, giving $x \in A \setminus (B \cup C)$. ■

7 SYMMETRIC DIFFERENCE

2 marks

Prove that $A \triangle B = (A \cup B) \setminus (A \cap B)$ for all sets A, B .

SOLUTION Recall $A \triangle B = (A \setminus B) \cup (B \setminus A)$.
 ($\subseteq$) Let $x \in A \triangle B$ be arbitrary, so $x \in A \setminus B$ or $x \in B \setminus A$. In the first case $x \in A \subseteq A \cup B$ and $x \notin B$, so $x \notin A \cap B$. The second case is symmetric. Either way $x \in (A \cup B) \setminus (A \cap B)$.
 ($\supseteq$) Let $x \in (A \cup B) \setminus (A \cap B)$ be arbitrary. Then x is in A or in B , but not in both. If $x \in A$ then $x \notin B$, so $x \in A \setminus B$; otherwise $x \in B$ and $x \notin A$, so $x \in B \setminus A$. Either way $x \in A \triangle B$. ■ "In at least one but not both" is the phrase that earns the second mark.

8 SUBSETS

2 marks

Prove that for all sets A, B : $A \subseteq B \iff A \cup B = B$.

SOLUTION ($\Rightarrow$) Suppose $A \subseteq B$. Certainly $B \subseteq A \cup B$. Conversely, let $x \in A \cup B$ be arbitrary: if $x \in B$ we are done, and if $x \in A$ then $x \in B$ by the hypothesis. So $A \cup B \subseteq B$, and the two sets are equal.
 ($\Leftarrow$) Suppose $A \cup B = B$. Let $x \in A$ be arbitrary. Then $x \in A \cup B$, which equals B , so $x \in B$. Hence $A \subseteq B$. ■ 1 mark per direction. This is an "if and only if", so two separate proofs are required – a single chain of $\Leftrightarrow$ signs scores at most 1.

9 CARTESIAN PRODUCT

2 marks

Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$ for all sets A, B, C .

SOLUTION Elements here are ordered pairs, so an arbitrary element is written (x, y) .
 ($\subseteq$) Let $(x, y) \in A \times (B \cap C)$ be arbitrary. Then $x \in A$ and $y \in B \cap C$, so $y \in B$ and $y \in C$. Hence $(x, y) \in A \times B$ and $(x, y) \in A \times C$.
 ($\supseteq$) Let $(x, y) \in (A \times B) \cap (A \times C)$ be arbitrary. Then $x \in A$, $y \in B$ and $y \in C$, so $y \in B \cap C$ and $(x, y) \in A \times (B \cap C)$. ■ 1 mark for taking an arbitrary *pair* rather than an arbitrary element, 1 for completing both directions.

Part C · Three to disprove, and one to write (6 marks)

10 DISPROOF

3 marks

Each statement below is **false**. For each, give explicit sets, compute both sides, and name an element that is in one side and not the other. One mark each.

1. $(A \cup B) \setminus B = A$ for all sets A, B .
2. $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ for all sets A, B .
3. $A \times B = B \times A$ for all sets A, B .

SOLUTION (a) $A = B = \{1\}$: left is $\{1\} \setminus \{1\} = \emptyset$, right is $\{1\}$. The element 1 is in the right and not the left. (It holds precisely when $A \cap B = \emptyset$.)
 (b) $A = \{1\}, B = \{2\}$: left is $\mathcal{P}(\{1, 2\})$, which contains $\{1, 2\}$; right is $\{\emptyset, \{1\}\} \cup \{\emptyset, \{2\}\}$, which does not. The element $\{1, 2\}$ separates them. (Only $\supseteq$ holds in general.)
 (c) $A = \{1\}, B = \{2\}$: left is $\{(1, 2)\}$, right is $\{(2, 1)\}$, and $(1, 2) \neq (2, 1)$ since ordered pairs agree only when both coordinates do. No mark for "it is false" without both sides computed. No mark for a Venn diagram.

11 FIND THE ERROR

2 marks

A student submits the following. Identify the first step that is wrong, say why it is wrong, and state whether the claimed identity is nevertheless true.

Claim. $A \setminus (B \setminus C) = (A \setminus B) \cup C$ for all sets A, B, C .

"Proof." Let $x \in A \setminus (B \setminus C)$. Then $x \in A$ and $x \notin B \setminus C$. Now $x \notin B \setminus C$ means $x \notin B$ or $x \in C$. So $x \in A \setminus B$ or $x \in C$, hence $x \in (A \setminus B) \cup C$. Therefore the two sets are equal. ■

SOLUTION The individual steps of the displayed argument are correct, and they establish $A \setminus (B \setminus C) \subseteq (A \setminus B) \cup C$. The error is the final sentence: only **one** inclusion was proved, and equality requires both. In fact the identity is **false** – take $A = \emptyset, C = \{1\}$: the left side is $\emptyset$ and the right side is $\{1\}$. 1 mark for locating the error at the conclusion rather than mid-argument, 1 mark for the counterexample. A student who says "the third step is wrong" has not read carefully and scores 0 for that part.

In no more than five sentences: a Venn diagram convinced you that some identity was true, and then you were asked to prove it by double inclusion anyway. What does the proof give you that the picture does not? You are not being asked to prove anything here.

MARKING

Look for the idea that the diagram shows one configuration while the claim quantifies over all sets, and/or that a diagram cannot be drawn faithfully for four or more sets. Credit any answer that distinguishes "I am convinced" from "it is established". There is no single correct answer.